# 4.1 Graphing Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.1 · Student sheet

**I can…**

- identify polynomial functions and give their degree, type, and leading coefficient.
- evaluate polynomial functions.
- describe the end behavior of polynomial functions and graph them.

**Vocabulary** · fill in the missing word

**Polynomial function** — $f(x) = a_n x^n + \dots + a_1 x + a_0$ with _____ exponents

**Degree** — the _____ exponent

**Leading coefficient** — the coefficient of the term with the _____ exponent (in standard form)

**End behavior** — _____

**Types** — _____ (0), linear (1), quadratic (2), cubic (3), quartic (4)

## Part 1 · End behavior sort

| Function | Degree (even/odd) | Leading coefficient (+/−) | Left end | Right end |
|---|---|---|---|---|
| $x^2$ | | | | |
| $x^3$ | | | | |
| $-x^2$ | | | | |
| $-x^3$ | | | | |
| $x^4$ | | | | |
| $-x^5$ | | | | |

Rule: _____

## Part 2 · Notes and examples (fill in each note, then work the example)

**Degree and leading coefficient:** Degree = the greatest _____. Leading coefficient = coefficient of the term with the _____.

### Example 1

Decide whether each function is a polynomial function. If it is, write it in standard form and give its degree, type, and leading coefficient.

a. $f(x) = 5x - 2x^3 - 8$ b. $g(x) = x^2 + 3x^{-1}$ c. $h(x) = 6 - \sqrt{2}\,x^4$ d. $k(x) = 5^x$

### Example 2

Evaluate $f(x) = x^4 - 3x^2 + 2x$ when $x = -2$.

**End behavior:** Even degree: ends go the _____ way. Odd degree: ends go _____ ways.

**Negative leading coefficient:** A negative leading coefficient _____ the ends.

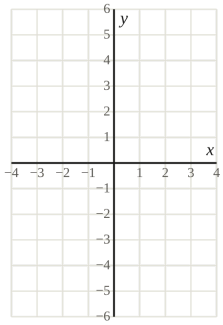
### Example 3

Describe the end behavior.

a. $f(x) = -0.5x^4 + 3x^2 - 1$ b. $g(x) = 2x^5 - x$

### Example 4

The graph shows a polynomial function. Is its degree even or odd? Is its leading coefficient positive or negative?




## Part 3 · Partner practice

### Partner 1

Write $f(x) = 4 - 7x + x^3$ in standard form. Give its degree, type, and leading coefficient.

### Partner 2

Describe the end behavior of $f(x) = -3x^3 + x$.